Toward Intelligent Intent-Based Network Slicing for IoT Systems: Enabling Technologies, Challenges, and Vision

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***Abstract*—The rapid integration of intelligence and automation into future Internet of Things (IoT) systems, empowered by Intent-based Networking (IBN) and Network Slicing (NS) technologies, is transforming the way novel services are envisioned and delivered. The automation capabilities of IBN depend significantly on key facilitators, including data management and resource management. A robust data management methodology is essential for leveraging large-scale data, encompassing service-specific and network-specific data, enabling IBN systems to extract insights and facilitate real-time decision-making. Another critical enabler involves deploying intent-based mechanisms within an NS system that translate and ensure user intents by mapping them to precise Management and Orchestration (MO) commands. Nevertheless, data management in IoT systems faces significant security and operational challenges due to the diverse range of services and technologies involved. Furthermore, intent-based resource management demands intelligent proactive, and adaptive MO mechanisms that can fulfill a wide range of intent requirements. Existing surveys within the field have focused on technology-specific advancements, often overlooking these challenges. In response, this paper defines Intelligent Intent-Based Network Slicing (I-IBNS) systems exemplifying the integration of intelligent IBN and NS for the MO of IoT systems. Furthermore, the paper surveys I-IBNS systems, focusing on two critical domains: resource management and data management. The resource management segment examines recent developments in IBN mechanisms within an NS system. Meanwhile, the second segment explores data management complexities within IoT networks. Moreover, the paper envisions the roles of intent, NS, and the IoT ecosystem, thereby laying the foundation for future research directions.**

***Index Terms*—Internet of Things, intent-based networking, network slicing, artificial intelligent, machine learning, resource management, data management.**

I. Introduction

INTERNET of Things (IoT) promises novel services such as autonomous vehicles, smart homes, and smart industries yet these innovations bring forth substantial Management and Orchestration (MO) challenges. Novel IoT services encompass a wide range of Quality of Service (QoS) requirements, including delay and throughput stressing the need for customized service deployments. Moreover, IoT networks are inherently heterogeneous, necessitating integration across various technological domains such as edge computing, access networks, and cloud infrastructure. To address these complexities, Network Slicing (NS) and Intent-based Networking (IBN) have emerged as promising solutions within the realm of network MO.

IBN emerges as a transformative technology with the primary goal of simplifying network MO through the automation of network operations based on overarching objectives. This methodology fundamentally distinguishes between the "what" and the "how." The "what" signifies the objective or desired outcome of a request, referred to as the intent, while the "how" pertains to the precise system configurations required to achieve that objective [1]. IBN accomplishes this separation by employing a middleware IBN system that serves as a crucial link between the user plane, where intents are initially received, and the network plane, where the actual intent fulfillment takes place. In addition, the IBN system employs key components, including intent translation which converts intents into MO policies, and intent assurance which monitors intent performance, promptly alerting the system to deviations from required performance levels. Those deviations often stem from changes in the network's state or user preferences and updates [2]. IBN components collectively enable a closed control loop of network resources that rely on an elastic MO paradigm facilitated by data-driven algorithms [3].

In addition, NS is a key technology for supporting the broad spectrum of IoT services. User intents within an IoT system can include functional requirements, such as mobility and firewall functions, and non-functional requirements, such as delay and security level [4], [5]. NS leverages softwarization and virtualization technologies to establish customized end-to-end service-oriented networks [6]. Consequently, NS facilitates the concurrent realization of multiple intents by enabling end-to-end network instances tailored to the functional and non-functional requirements of the intents [4].

Moreover, the integration of Artificial Intelligence (AI) and Machine Learning (ML) algorithms into IBN systems emerges as a promising avenue [7] to autonomously support intent requirements. These advanced algorithms exhibit the capability to discern the real-time state of the network, enabling them to make informed decisions aimed at optimizing both network and service performance [8]. For instance, Reinforcement

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Learning (RL) algorithms have proven superior performance in optimizing resource allocation policies in different network domains, such as power and radio resource allocation in the access domain [9], and virtual computation and storage resource allocation in the edge domain [4]. Those policies can be enforced by the IBN system to support autonomous network operations that adhere to intent requirements [10].

These technologies have the potential to enable IoT systems to realize novel services while addressing fundamental MO challenges. However, the development of advanced IoT systems remains in its infancy, necessitating further in-depth examination. To this end, we define an Intelligent Intent-Based Network Slicing (I-IBNS) system to be a MO system that facilitates IBN, employs NS to manage its resources, and integrates AI/ML to optimize its operations. In this paper, we focus on two fundamental enablers of I-IBNS systems by conducting a comprehensive literature review of existing work on resource management and data management. This review discusses both the potential and limitations of current research in meeting the requirements and addressing the challenges associated with I-IBNS systems. Following the literature survey, we present a forward-looking perspective on IoT systems, defining the roles of intent, NS, and the broader IoT ecosystem, hence, providing a foundation for future research directions.

A. Contributions

The primary contributions made in this paper can be summarized as follows:

- *Survey of I-IBNS Systems:* A thorough examination of I-IBNS systems is undertaken, involving the related literature of two crucial domains: resource management and data management. The exploration of the resource management aspect encompasses an evaluation of the literature on intent-oriented NS, while the data management aspect focuses on mechanisms employed for data management within IoT systems. This comprehensive survey not only consolidates existing knowledge but also identifies requirements, gaps, and opportunities for further research.
- *Future IoT Systems:* A prospective outlook on future IoT systems is provided, specifically concentrating on clarifying the roles of intent, NS, and the broader IoT ecosystem. The comprehensive overview serves to elucidate the terminologies associated with intent and NS, outlining their evolution within the context of upcoming IoT systems. Furthermore, it highlights the trajectory of future IoT implementations, emphasizing their dependence on an open and interconnected ecosystem.

The paper is organized into six sections with the core sections shown in Figure 1. A comparison to related surveys is presented in Section II. A background on IBN, NS, and related technologies is presented in Section III. Section IV includes a comprehensive survey on I-IBNS for IoT systems. An overview of future IoT systems is presented in Section V. Section VI discusses the research directions. Lastly, the conclusion is in Section VII.

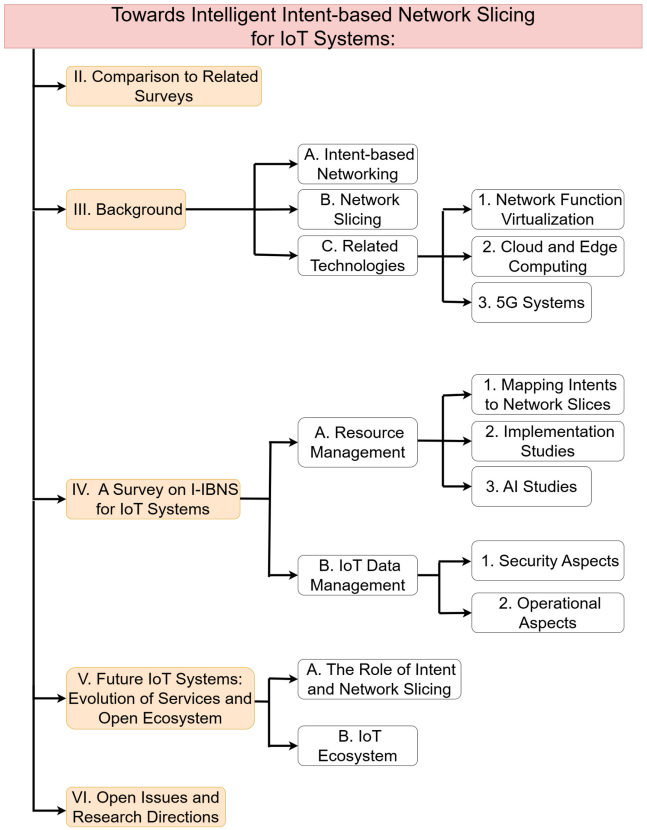


Fig. 1. Outline of the Main Sections of the paper.

II. Comparison to Related Surveys

The expanding body of literature in the realms of IBN and NS has garnered significant attention. In [2], the authors discuss the fundamental constituents comprising a closed-loop intent automation system, encompassing intent translation, resolution, activation, and assurance. The paper surveys the operational mechanisms of IBN components, delving into recent literature that expounds on each facet. In a related vein, [11] underscores the standardization initiatives and industrial endeavors within Intent-Driven Networks (IDN), emphasizing architectural considerations. A concise survey in [12] provides insights into standardization and platforms underpinning intent-based systems. Additionally, [13] introduces the concept of intent refinement, a method for transforming declarative intents into machine-readable policies. The paper reviews existing intent refinement schemes and proffers an intelligent intent refinement system based on natural language processing.

Concurrently, NS has been the subject of extensive examination, with surveys encapsulating various NS aspects. In [14], the authors discuss the functionality of end-to-end NS and survey NS modeling works across Radio Access Networks (RAN), Core Networks (CN), and Transport Networks (TN). The paper also discusses the lifecycle management of NS, spanning from service request to deployment. Focusing on cloud-native 5G NS, [15] explores Mobile Edge Computing (MEC) and NS, discussing their role in 5G service-centric use cases. The survey in [6] conducts a comprehensive analysis of NS in the context of IoT, addressing challenges in various IoT applications and scenarios. Similarly, [16] explores NS for IoT

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applications, with a specific emphasis on satellite-terrestrial edge computing networks. The paper delves into slice MO across diverse satellite integration architectures.

Numerous surveys probe the infusion of AI and ML into NS systems. Reference [17] discusses NS management in Industrial IoT (IIoT) services, encompassing domains such as smart transportation, smart energy, and smart factories. The paper proposes an architecture for intelligent NS management tailored for IIoT. Reference [18] delivers a comprehensive survey on the contributions of ML in NS, identifying major challenges in resource allocation, NS admission, and traffic forecasting and classification. Meanwhile, [19] centers on the scaling and placement of the Virtual Network Function (VNF) chain within a network slice, surveying various Deep Reinforcement Learning (DRL) approaches. The paper discusses diverse DRL models and their efficacy in addressing specific facets of the problem. Finally, [20] surveys AI-enabled NS across all network segments, organizing the literature in alignment with the phases of the NS lifecycle.

To the best of the authors' knowledge, no comprehensive work has been undertaken to furnish a taxonomy and clarify the current trajectory of I-IBNS systems. This study defines I-IBNS systems, discusses their challenges, and conducts a thorough literature review to identify gaps and future research directions. The aim is to accelerate research in the advancement of I-IBNS systems by presenting the reader with comprehensive literature in relevant domains. Additionally, the study seeks to establish a forward-looking perspective on future IoT systems including the evolution of intent and the IoT ecosystem.

III. Background

This section presents an overview of the IBN and NS technologies as they are the main components of the I-IBNS system. Moreover, a background on technologies related to the context of this work, such as Network Function Virtualization (NFV), cloud, edge, and 5G systems, is provided.

A. Intent-Based Networking: The Advent of Autonomous Networks

IBN is a new network automation technology that has attracted significant attention from research and standardization bodies. In this section, an overview of IBN is provided along with its related definitions and enabling components.

IBN Evolution: There is a consensus among research and industry bodies that traditional manual configuration is no longer a practical solution. This is due to the major changes in the way communication and computing systems operate, which significantly increased the complexities of system management and operations [21]. Those challenges include the requirement to support multi-tenancy environments with heterogeneous technologies. IBM was the first to introduce an innovative concept based on the Monitor, Analyze, Plan, and Execute (MAPE) loop to enable network automation [22].

The field of autonomous network management has undergone multiple phases, one of which is policy-based network management (PBNM) developed by the Internet Engineering Task Force (IETF) [23]. Particularly, PBNM requires high-level management policies to automate the systems' configurations. Those policies define a set of control loops that map the events and conditions to corresponding actions. A PBNM system is expected to decompose the high-level management policies into low-level configuration commands.

Software Defined Networking (SDN) has revolutionized the field of autonomous network management by enabling centralized control of networks [24]. This is achieved through SDN controllers that support the programmability of the infrastructure. The advantage of SDN lies in the simplification of the automation of system configurations. This is made possible through the separation of the control plane from the data plane, which allows for the centralized management of network resources. The programmability of the infrastructure further enhances the automation of system configurations, making it possible to easily adapt to changing network requirements.

IBN, the emergence of autonomous networks, facilitates the utilization of a declarative syntax, denoted as "intent," as the input mechanism, prioritizing a targeted result during system configuration. This methodology empowers the IBN system to manage autonomous system configurations through the enforcement of policy rules, guaranteeing the effective realization of specified intents. Through a focus on the outcomes of system configurations, the implemented policy rules within the IBN system ascertain that the network is systematically configured to align with the precise requirements articulated by the intents.

IBN Definition and Components: There has been a concerted effort to define and standardize the concept of intent. The IETF has provided a definition of intent as "an abstracted, declarative and vendor-agnostic set of rules used to provide full life cycle (Design/Build/Deploy/Validate) to a network and services it provides" [1]. Meanwhile, IBN has five key components that enable closed-loop automation of intent through its life cycle, namely, intent profiling, intent translation, intent resolution, intent activation, and intent assurance [2].

The process of IBN begins with a user expressing intent in a declarative form. This intent is then processed by an IBN system, which consists of several key components. The first of these is intent profiling, which accepts the declarative form request through a user-friendly interface such as natural language expressions or drop-down menus. The second component is intent translation, which acts as middleware between the user plane and the system plane. This component is responsible for translating the abstracted intent into network policy that can be applied to the system as configuration commands. The third component is intent resolution, which involves resolving conflicting intents before activating a new incoming intent. This stage is necessary as users can submit their intents unaware of already existing system configurations. Moreover, altering system configurations could affect the performance of intents in the system. The fourth component is intent activation. After the IBN system ensures a conflict-free intent policy, the intent activation proceeds with provisioning

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the resources required to realize the intent as well as the policies that ensure the intent performance. The fifth component is intent assurance which is responsible for continuously monitoring the performance of the intent. This is required because changes in the system could affect the performance of the intent. In such scenarios, the intent assurance should alert the IBN system to refine the policy or take corrective actions that ensure the successful delivery of the intents in the system.

B. Network Slicing

NS technology represents an inventive solution enabling systems to discern the distinct requisites of services and facilitate the concurrent deployment of multiple service-based logical networks within a shared physical infrastructure, thereby optimizing cost-effectiveness in service implementation. A noteworthy advancement within NS lies in its capacity to govern intricate systems by establishing a hierarchical structure of management layers. This section delineates the conceptual framework of NS, associated concepts and service model.

NS Definitions and Service Models: NS allows multiple logically separated networks to co-exist over the same infrastructure for serving specific purposes, service categories, or individual users. NS is defined to be a set of network functions (NFs) that communicate through virtual networks such as a Virtual Private Network (VPN) and a Virtual Local Area Network (VLAN) [14]. A Network Slice Instance *(NSI)* describes the active form of an NS that includes a logical network connecting a set of NFs and their resources. To ease the management of NS-based systems, the Third Generation Partnership Project (3GPP) identifies the NS subsystem *(NSS)* to be a subset of the entire NS [25]. A *NSS* can be either isolated, e.g., serves one specific NS, or shared between multiple NSs.

The NS service model comprises two principal categories: the Network Slice as a Service (NSaaS) model and the network slice service request. The NSaaS model represents a comprehensive virtual network, commonly sought after by vertical industries such as healthcare and smart cities. In this context, a formalized agreement, known as a Service Level Agreement (SLA), is established between the vertical industry and the system owner. This SLA delineates the Quality of Service (QoS) parameters governing the virtual network. Moreover, the arrangement may incorporate a Management Level Agreement (MLA), stipulating the MO capabilities delegated by the IoT system owner to the vertical industry, thereby providing an elevated degree of control over the virtual network. Notably, the NSaaS model draws parallels to the Infrastructure as a Service (IaaS) model within the domain of cloud computing.

The second category, the network slice service request, empowers end-users, including service consumers, to solicit access to services facilitated by network slices. In response, the system owner has the capability to generate multiple network slices, each tailored to deliver a distinct service. It is important to note, however, that in this mode of request, the users of the network slice do not possess direct control over its configuration or functionalities.

NS Management Layers: 3GPP has defined three management layers of NS systems [25]. The first layer is the Communication Service Management Function (CSMF), which accepts service requests and translates them to network slice configurations. This translation process can utilize a Network Slice Descriptor (NSD) [2], which provides configuration templates for implementing network slices. The following is the Network Slice Management Function (NSMF), which is responsible for the MO of Network Slice Instances *(NSIs)*. In addition, the NSMF derives NSS requirements from the network slice requirements. Lastly, the Network Slice Subsystem Management Function (NSSMF) communicates with the NSMF and is responsible for the MO of Network Slice Subsystem Instances *NSSI*. The *NSI*s, and *NSSI*s form the active state of network slices and network slice subsystems, accordingly.

C. Related Technologies

The full potentials of IBN and NS technologies are realized when utilizing virtualization, softwarization, and emerging communication systems. This section overviews NFV, cloud and edge networks, and 5G systems.

1) Network Function Virtualization (NFV): NFV is a technology that enables the deployment of network functions in virtual environments [26]. This is achieved by using Virtual Network Functions (VNFs), which are software implementations of network functions, and deploying them on Virtual Machines (VMs). By doing so, VNFs located in one or several cloud environments can be chained together to create services. The NFV Infrastructure (NFVI) represents the foundational hardware resources, such as compute, storage, and network, and software environment, such as virtual Central Processing Units (CPU) and memory, in which VNFs are deployed, managed, and executed. This infrastructure provides a flexible way of chaining VNFs using a Network Function Forwarding Graph (NFFG).

The European Telecommunications Standards Institute (ETSI) has standardized the Management and Orchestration (MANO) framework for managing and orchestrating VNFs over the NFVI [27]. The framework consists of three main components: the Virtualized Infrastructure Manager (VIM), the Virtual Network Function Manager (VNFM), and the NFV Orchestrator (NFVO). The VIM is responsible for managing the NFVI resources, while the VNFM is responsible for managing the VNFs. The NFVO is responsible for coordinating and managing the VIM and VNFM to ensure the efficient deployment and management of VNFs.

It is important to clarify that this paper uses the term MO to refer to management and orchestration systems in a general sense. In contrast, the term MANO specifically denotes the management and orchestration framework defined by ETSI.

2) Cloud and Edge Computing: Cloud computing and edge computing are two distinct paradigms that enable the provision of resources and services to users [28]. Cloud computing leverages virtualization technologies to offer on-demand storage, computational, and networking resources. These resources are typically provided through data centers

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distributed over multiple geographical locations. In contrast, edge computing enables applications to perform computations, service acquisition, and data analytics close to users, thereby reducing latency and improving bandwidth.

The MEC framework, presented by the ETSI, is a realization of edge computing that allows third parties to orchestrate various services on the edge layer [29]. This framework enables the deployment of a wide range of services, including augmented reality, location-based services, and video analytics, among others.

Fog computing, originally introduced by Cisco, represents a prominent realization of edge computing [30]. It facilitates efficient data communication and processing by extending cloud capabilities closer to the data sources. The fog computing paradigm adopts a hierarchical architecture that distributes computing, storage, and networking resources across multiple layers, spanning from the edge of the network to centralized cloud servers, thereby reducing latency, conserving bandwidth, and enabling real-time decision-making.

3) 5G Systems: 5G systems introduce a range of innovative features that facilitate rapid system integration and enhance operational efficiency [31]. Among these advancements, NS emerges as a prominent feature conceptualized within the framework of 5G networks. The 5G system incorporates essential functionalities like the Network Slice Selection Function (NSSF), facilitating device registration to network slices [14]. Another notable addition to the 5G landscape is the Network Data Analytics Function (NWDAF) [32], [33]. The NWDAF aggregates data from diverse 5G functions, including the Policy and Charging Rule Function (PCRF), Access and Mobility Function (AMF), Unified Data Management (UDM), and NSSF. Importantly, the NWDAF harnesses ML models to dynamically optimize network performance, encompassing tasks such as slice traffic prediction, resource utilization prediction, anomaly detection, and more.

A fundamental characteristic of 5G functions is their software-based implementation and design as cloud-native entities, bolstering network flexibility and scalability [31]. Moreover, these functions are crafted to operate seamlessly within an all-IP network environment, ensuring compatibility with the IP protocol stack. This design fosters smooth integration with external networks, including the edge and cloud.

In the domain of access management, RAN slicing has attracted considerable attention for its ability to partition radio resources, allowing for tailored service performance [34]. Additionally, 5G systems are poised to support a myriad of access technologies [31]. For example, the 5G core network functions are adept at managing incoming traffic from heterogeneous access systems, such as WiFi. This flexibility affords a diverse array of network design options capable of accommodating various service requirements.

IV. A Survey on I-IBNS for IoT Systems

This section provides a comprehensive survey on I-IBNS systems by reviewing the literature on two fundamental areas: resource management and data management. The resource management part reviews the literature of the intent-oriented NS mechanisms. Moreover, the data management section reviews the work related to IoT data management in I-IBNS systems.

A. Resource Management

NS has become a fundamental component in resource management frameworks, catering to diverse intent requirements within the context of I-IBNS systems. An overview of research work in this domain is presented in Table I, comparing technology domains, intent-to-network slice mappings, and primary focus areas of each study. While not all studies integrate AI/ML mechanisms, NS consistently emerges as the primary technology for resource management across the literature, emphasizing one or more of three central aspects: intent-to-network slice mapping, implementation considerations, and AI/ML integration. In the following, we examine each of the three aspects in detail. After each discussion, we highlight key insights and reflections specific to the respective aspect.

1) Mapping Intents to NSs: In the process of mapping intents to network slices within I-IBNS systems, the literature explores diverse methodologies, including mapping to network slice types, templates, or specific parameters like VNFs, NFFGs, or resource descriptors. For instance, [42] employs an approach that identifies conceivable network slice solutions based on tenant intents, concurrently incorporating operator policies. This approach underscores the interplay between user intent and operational policies in shaping network slice configurations. Another study, [5], addresses the challenges of ensuring end-to-end NS services across multiple verticals. The work relies on intent classification into business, service, and resource types, providing a structured foundation for autonomous intent-driven management.

Key Insights: In the context of mapping intents to NSs, there exists a clear trade-off between the complexity of implementation and the breadth of capabilities that can be accommodated. For instance, adopting predefined NS templates for intent mapping limits users to a predefined set of service types. This approach greatly streamlines the automation of service deployment, allowing near-instantaneous mapping of user intents to these templates. However, to fully leverage the capabilities of I-IBNS, more agile approaches are required. These approaches should support the creation of a fully customized service without being constrained by predefined templates. This flexibility enables users to express intents in a broader, more generalized manner, necessitating the I-IBNS system to design NS capable of meeting specific user expectations.

We refer to the range of flexibility with which I-IBNS systems can incorporate intents as the intent space. Most approaches in the literature, as exemplified in Table I, map intents to pre-defined NS templates or service types, thereby supporting a narrow intent space. Further research is needed to develop methodologies that support broader intent spaces, allowing users to express intents in an unrestricted manner. For instance, the advancement of Large Language Models (LLMs)

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TABLE I
RELATED WORK ON I-IBNS SYSTEMS

Work	Year	Domain	Intent to Network Slice Mapping	Focus
[36]	2020	SDN and NFV	Service type to policy catalogs	IBN testbed implementation
[37]	2020	RAN and Core	Service requirements to NFFG	Resource scaling-based on prediction
[38]	2021	RAN and Core	QoS to eMBB, URLLC slice templates	Resource scaling-based on prediction
[4]	2021	RAN and Core	QoS to eMBB, URLLC slice templates	Resource scaling-based on prediction
[39]	2021	SDN and NFV	Service requirements to intent type	NS deployment implementation
[40]	2021	NFV	Service profile to network slice template	Smart grid NS deployment framework
[41]	2021	SD-WAN	Network intent descriptor	NS deployment in SD-WAN Orchestrator
[42]	2021	5G	Select network slice type	NS deployment framework for vertical industry
[43]	2021	Access and Core	Select network slice type	NS design considering operator policies
[44]	2021	RAN and Core	Resource description framework	NS deployment framework enforcing CLC policies
[10]	2021	NFV	Intents to slices templates with VNFs	Decomposing and placement of intent VNFs
[45]	2021	5G in Industry 4.0	Slice parameters to customize NST	Design and implementation of intent NS
[46]	2022	RAN	Network slice mapping using data models	Architecture and implementation of data-driven IBN
[47]	2022	RAN and Core	Select network slice type	Resource utilization prediction
[8]	2022	RAN	QoS templates	DQN for policy optimization
[48]	2023	RAN and Core	User intent to NFFG	AI resource prediction
[10]	2023	RAN	Select network slice type	DQN for policy optimization
[5]	2023	RAN and Core	Business and service intent to resource intents	Architecture and implementation of intent NS
[49]	2023	Private 5G	Slice selection model	Design and implementation of intent NS
[50]	2024	RAN	Intent to QoS requirements	Resource allocation based on intent decomposition

can be employed to extract NS configuration parameters by communicating with users [50]. Subsequently, the system can autonomously design and deploy customized NSs within the network infrastructure. The process of extracting NS design configurations can involve iterative communication with the user to fill in any remaining gaps in NS configuration.

2) Implementation Studies: Several studies emphasize implementation aspects, validating emerging technologies' capabilities in automating fundamental network configuration tasks. These automation mechanisms leverage policy catalogs [35], NFV orchestrators [38], [48], knowledge management tools [5], and adaptive policy execution engines [43].

Key Insights: While implementation studies have focused primarily on automating network slice deployment, they have largely overlooked other critical IBN components, particularly intent assurance. This component is essential for monitoring network slice performance throughout its lifecycle and adapting to dynamic network conditions. Future research could explore the integration of ML-based policies for intent assurance, with empirical validation in real-world network environments to demonstrate their effectiveness and reliability.

3) AI Studies: Incorporating AI/ML algorithms within the I-IBNS literature optimizes resource allocation while considering intent requirements and system dynamics. In [36], a topology-aware Graph Neural Network (GNN) enhances VNF resource scalability, enabling proactive resource management. The use of Long-Short-Term Memory (LSTM) networks in 5G NWDAF [37] predicts slice load, user mobility, traffic forecasts, and anomalies, supporting the I-IBN system in meeting intent requirements. Hybrid ensemble learning (EL) algorithms are proposed for network resource utilization prediction, anomaly detection, and mitigation in 5G NWDAF [46] [47], showcasing a diverse range of AI/ML contributions. Furthermore, the work in [38] delves into resource management and orchestration within multi-domain slices. It introduces an intelligent mechanism based on deep learning, specifically designed to address run-time resource scalability and assurance. Moreover, the work in [51] proposes a Deep Reinforcement Learning (DRL)-based system that ensures end-to-end intents with specific delay requirements. Another DRL-based approach is proposed in [52], which considers the system costs, including Service Level Agreement (SLA) and over-provisioning costs in deploying intents on edge-enabled IoT systems.

In the RAN domain, an SDN controller with a Deep Q-Network (DQN), a foundational Reinforcement Deep Learning algorithm, is proposed [8]. This controller actively collects real-time QoS metrics, encompassing network delay, actual throughput, and resource utilization, and employs the DQN-Greedy algorithm to optimize slice configurations for radio resource block allocation. In another study, [10], an intent-driven multi-agent Deep Q-Network (MA-DQN) algorithm is introduced to maximize SLA satisfaction through radio block resource allocation, further exemplifying the integration of AI/ML in addressing intent-driven requirements.

Key Insights: While the research area on AI/ML for resource management in I-IBNS is highly active, many challenges remain unaddressed. Importantly, to the best of the authors' knowledge, no existing work has tackled a comprehensive I-IBNS system that integrates the intent translation mechanism from the user domain with intent activation and assurance in the network domain. Most research efforts have concentrated on one of these domains without addressing the complexities of integrating both aspects. Typically, AI/ML resource management research has assumed the support of various NS types exemplifying intent scenarios. However, further investigation is needed into the intelligent mapping of intents to NSs and the associated intent assurance policies. Comparing different approaches to intent mapping and assurance within end-to-end systems could provide valuable insights.

Additionally, there is a significant research gap in the implementation of AI/ML resource management for I-IBNS systems. The practical aspects of deploying these advanced techniques in real-world scenarios require more thorough exploration and experimentation, especially, clarifying aspects such as the response time of proactive policies.

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B. IoT Data Management

Data management within I-IBNS systems assumes a paramount role in facilitating informed MO decisions throughout the system's operational spectrum. More specifically, the data management framework is tasked with routing and aggregating extensive, multi-sourced data originating from various network segments and presenting this data to MO entities.

The encompassed network data comprises system-specific information, including system utilization metrics, and service-centric performance data, such as latency and reliability. For example, transport networks encompass crucial physical network details, such as link capacities, physical topological attributes, and the number of routers, among others. Furthermore, transport networks diligently monitor various Key Performance Indicators (KPIs), such as packet drop rates and latency metrics. In parallel, the sensor plane actively manages the aggregation of sensor data, efficiently routing them to designated NSIs, and furnishes reports detailing the capabilities of sensor nodes, ranging from gas sensors to cameras. For instance, in [53], the study collected real thermal video data from a liquefied natural gas (LNG) facility, focusing on leak detection scenarios. The collected data relies on thermal cameras and is essential for context-aware sensor placement. Other data types include user-related data such as mobility patterns and technology-related activities.

Leveraging this wealth of data, MO entities, such as the CSMF, NSMF, NSSMF, and intra-domain entities such as NFVO and SDN controllers, can engage in data analytics to facilitate advanced decision-making in different intent-related tasks. The sensor information repository empowers the I-IBNS management layers to make informed determinations regarding the suitability of the deployed sensors in fulfilling the specific requirements of a given intent [54]. In addition, user behavior data such as mobility [55], and technology-related activities [8] can be utilized by the I-IBNS system for resource management. Data analytics techniques may be harnessed for modeling network or service performance, thereby supporting proactive orchestration initiatives [37]. Furthermore, the ongoing monitoring of deployed service performance serves to ensure compliance with stipulated QoS benchmarks, fostering a responsive and adaptive system [10].

The following discussion centers on the challenges associated with data management frameworks, addressing the security and operational aspects, as summarized by Fig. 2.

1) Security Aspects: The establishment of an effective data management framework necessitates an examination of security-related concerns that may manifest across diverse operational domains. In the following, we examine each of the security aspects in detail. After each discussion, we highlight key insights and reflections specific to the respective aspect.

- *Sensor Data Security:* In the literature, there exists a notable emphasis on the security of the sensor layer, encompassing data encryption, device identity management, authentication mechanisms, access control, trust management, and intrusion detection and prevention systems [56]. The sensor domain within IoT presents a set of formidable challenges, characterized by the imperative need to implement security mechanisms on resource-constrained devices endowed with heterogeneous capabilities, dynamic behavioral patterns, and scalability issues.
  Addressing these challenges has prompted the proposition of adaptive measures such as adaptive data transmission [57] and cluster head selection mechanisms [58]. The former seeks to value information presented in IoT data for optimizing transmission decisions, thus ensuring efficient energy management of IoT sensor nodes. Meanwhile, cluster head selection mechanisms aim to enhance security through strategic delegation of responsibilities within IoT networks, thereby mitigating the impact of resource limitations and fostering scalability.
  Moreover, emerging technologies such as edge computing and blockchain have garnered significant attention as potential enablers of enhanced security paradigms. Edge computing is explored for accommodating various tasks, such as data management [59], intrusion detection and prevention systems [60], [61], and trust management [62]. Moreover, the integration of blockchain technology is being actively investigated for its potential in device management, authentication, and access control, among others, providing a decentralized and tamper-resistant framework that augments the security posture of IoT ecosystems [63], [64], [65].
  Key Insights: Future work could focus on developing holistic data security frameworks that integrate aforementioned technologies and address the requirements of I-IBNS systems. For instance, the integration of IoT security with I-IBNS automation presents a delicate balance between protection and performance. While robust security measures are essential for protecting intent-based operations, security overhead could impact system responsiveness and resource availability. This requires developing context-aware security solutions that can effectively protect NS intent requirements without compromising the system's ability to rapidly respond to changing network conditions.
- *System Data Security:* A further security consideration pertains to the safeguarding of system data, an imperative arising from the multi-tenancy characteristics inherent in virtual environments. System data encapsulates KPI metrics, such as execution times, alongside physical parameters such as electromagnetic emission and supply current. In I-IBNS systems, tenant network slices can observe the system data, thereby introducing potential vulnerabilities that may be exploited to compromise shared resources or compromise the privacy of cryptographic operations, leading to the extraction of secret keys, known as side-channel attacks [66], [67].
  Security concerns in multi-tenancy NS environments have been addressed through slice isolation [68]. However, it is essential to underscore that while NS isolation enhances the security posture of the network slice, it concurrently impacts resource sharing and utilization in the system. Consequently, MO systems can optimize the NS isolation

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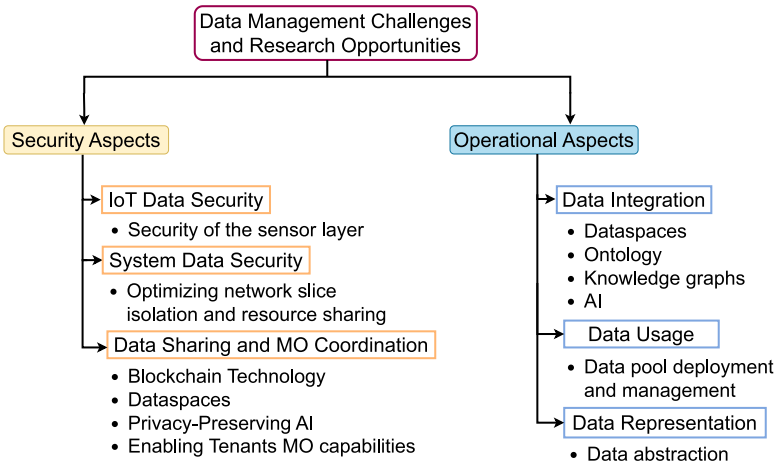


Fig. 2. Data Management Challenges and Research Opportunities in I-IBNS IoT Systems.

and resource sharing [69]. Moreover, NS isolation can take place within one or more network segments, e.g., RAN domain or edge domain, to increase the utilization of shared network resources [68].

Moreover, in I-IBNS systems, the required security level can be expressed by the user intent or included as part of the QoS requirements [4], [37]. Accordingly, the I-IBNS system is expected to translate the user intent into security policies, which can include data encryption mechanisms, access control, and NS isolation requirements. Those security policies can be enforced by the different system domains [43].

Key Insights: The protection of system data in I-IBNS environments requires considering the automation behavior of IBN that are not addressed by the literature. A key challenge is balancing network slice isolation with efficient resource utilization while maintaining the system's ability to fulfill intent requirements across multiple coexisting network slices. For instance, when resource demands fluctuate, e.g., when one network slice requires additional resources, the I-IBNS system must dynamically adjust resource allocation to maintain specific intent requirements such as delay and throughput guarantees. These dynamic resource management policies must be carefully optimized to maintain appropriate levels of NS isolation without compromising performance.

- *Data Sharing Security:* Lastly, the security of data sharing and MO coordination is of paramount significance to I-IBNS systems. For instance, in IoT systems with multi-administrator domains, the necessity for secure data sharing and MO coordination is imperative to facilitate seamless end-to-end network implementations. Moreover, utilizing the NSaaS model allows system owners to offer network slices to tenants with MO control capabilities, which requires security considerations. The data subject to sharing spans a wide spectrum of information, encompassing communication service performance metrics, system performance metric,s including resource utilization statistics, and essential physical system attributes, such as the geographical coordinates of system resources. In this scenario, the security of data-sharing between entities belonging to different organizations is of paramount importance, encompassing possible vulnerabilities such as the trust of legal data use and data leaks that can occur during the sharing process or within the tenants' systems [70].

Numerous proposals have emerged in the literature as proactive measures to address the security concerns surrounding data sharing and MO coordination. A pivotal proposition involves the integration of blockchain technology to fortify data sharing mechanisms, e.g., in multi-domain [71], and multi-cloud [72] environments. Blockchain, renowned for its decentralized and tamper-resistant nature, introduces a secure and transparent layer to the data-sharing process. This ensures the integrity of shared information and establishes an auditable trail of transactions, mitigating the risk of unauthorized access and tampering. Additionally, advanced data analytics can be utilized by MO systems, by which only service or traffic attributes are shared, therefore addressing the privacy constraints of data sharing [73], [74].

An additional avenue for enhancing data management security arises from the realm of data management systems, notably within the frameworks of data ecosystems and dataspaces. A data ecosystem is characterized by a collective of multiple actors, wherein each participant assumes roles such as data consumers, producers, or providers [75]. In I-IBNS systems, the composition of a data ecosystem may encompass entities such as the system owner, NSaaS providers, and system users. This collaborative arrangement encapsulates a diverse spectrum of stakeholders, each contributing to the data ecosystem's dynamic fabric [76].

Furthermore, dataspaces emerge as a strategic architectural approach designed to navigate the complexities of decentralized structures inherent in data ecosystems [77]. Dataspaces can facilitate the systematic management of stakeholders and their intricate interrelations within the broader data ecosystem. This architectural paradigm is particularly salient in orchestrating and optimizing

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data flows, ensuring that disparate entities within the ecosystem can seamlessly collaborate while maintaining the security of shared data through agreements and policy enforcement [78].

Privacy-preserving AI techniques have also garnered attention in mitigating security concerns within the context of MO coordination. Federated reinforcement learning [79] and knowledge transfer [80], [81] approaches, recognized as privacy-preserving AI paradigms, facilitate collaborative learning without the necessity of centralizing sensitive data. These techniques empower orchestrators to collectively improve their models by collaboratively training on local datasets without sharing raw data, thereby addressing privacy concerns and promoting secure MO coordination across diverse orchestrators.

Another proposal is to allow the extension of MO capabilities to tenants [82]. In this framework, MO agreements dictate the allocation of special interfaces to tenants, enabling them to autonomously manage and orchestrate their virtual resources. Such proposals rely on the MANO-to-MANO interfaces in supporting a collaborative MO between entities [83], [84], wherein the security of the MO coordination is enforced by security level agreements.

Key Insights: While the research has focused on policy enforcement for secure data sharing or advanced technologies such as blockchain, an additional layer of security mechanisms can fortify the security posture of the I-IBNS system. For instance, selective data sharing could be a possible venue where sophisticated approaches are deployed to balance operational effectiveness with data protection. Moreover, rather than sharing raw system data, advanced analytics can process and abstract essential information, for example, sharing bandwidth utilization trends rather than detailed network traffic data, or aggregated performance metrics instead of granular system measurements. The sharing of abstracted data could help protect sensitive information while still enabling effective coordination and resource optimization. Nevertheless, more research is needed to study the trade-off between data abstraction and the efficiency of resource management systems.

Moreover, although system orchestration security has been studied in the NFV domain, NS orchestrator security is still an open issue [68]. The security consideration in NS orchestration is crucial, especially in I-IBNS systems that allow the extension of MO capabilities to tenants.

2) Operational Aspects: Concerning the operational complexities of data management frameworks, a constellation of challenges exists. These challenges are centered on aspects such as data integration, data usage methodology, and data representation. Each of these challenges carries distinct significance in the MO of I-IBNS systems.

- *Data Integration:* Data integration is essential as resource management algorithms grapple with integrating heterogeneous data from various technological systems [85]. In I-IBNS systems, the NSMF can manage NFV and RAN domains. Consequently, the resource management algorithms of NSMF must process data received from these entities to optimize end-to-end service objectives. The orchestration of such diverse data streams hinges on robust data integration methodologies.

  Numerous scholarly endeavors have been undertaken to tackle the challenges associated with data integration, including the utilization of dataspace, ontology, and knowledge graphs. Dataspace proves instrumental in assimilating diverse data types originating from disparate systems, enabling seamless integration using data management systems [86]. Additionally, ontology-based methods contribute to enhancing data integration by establishing a shared understanding of the domain's semantics. This can be achieved using standardized data models, such as the Resource Description Framework (RDF) which provides a formalized and flexible representation, fostering interoperability and compatibility across heterogeneous systems [87]. Moreover, knowledge graphs offer a structured representation of information and relationships, facilitating effective integration. Knowledge graph has been used to integrate network status data and intent requirement [88]. The integration of the intent requirement and network status data aids efficient mapping of the unified representation into network policies.

  Another avenue for data integration is the use of AI/ML algorithms. For instance, in [79], federated RL is proposed to coordinate multiple heterogeneous domains in RAN systems. In such an approach, federated learning is utilized to allow intra-domain learning and sharing of those ML models with a centralized coordinator/orchestrator. Moreover, Vertical Federated Reinforcement Learning (VRL) allows collaboration between heterogeneous systems. This methodology can be expanded in I-IBNS systems, such that NSSMFs utilize VRL to train their local models and share those models with the NSMF which will integrate the shared knowledge and aid in end-to-end MO collaboration.

  Key Insights: In I-IBNS systems, there remain important considerations that are limited in the literature, such as scalability and time synchronization of data integration. As the system scales to handle more network slices and domains (RAN, NFV, Edge), the integration solution must efficiently process growing volumes of heterogeneous data while maintaining real-time operation capabilities. Time synchronization becomes critical since NSMF must coordinate data from multiple NSSMFs operating at different timescales, for instance, RAN domain data may be collected at millisecond intervals, while NFV domain metrics might be gathered at longer intervals. Misaligned timestamps can lead to incorrect resource allocation decisions and QoS degradation. There is a need for studying the efficiency of data integration solutions in aligning multi-sourced data while scaling to handle increased data volumes.
- *Data Usage:* Data usage methodology is tasked with unraveling the complexities of data utilization across various management systems. For instance, the data

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generated by an NSSMF might find utility in multiple NSMFs. Additionally, the management functions and resource optimization algorithms of the NSMFs may be hosted on disparate physical servers, lending a degree of complexity to the scenario. To navigate this intricacy, a data management framework may institute a central data pool, serving as a hub for aggregating multi-sourced big data [37], [46]. To regulate access and utilization, specialized protocols come into play, governing the management functions' interaction with this data pool. These protocols can facilitate data inquiries from the MO entities [89].

Key Insights: While existing literature has explored the issue of data utilization among MO entities within different multi-administrative domains [71], [72], there exists a need for the adoption of a more thorough and strategically oriented approach for data usage. For instance, with the reliance on AI/ML in I-IBNS systems, data management frameworks must be agile enough to accommodate scenarios where MO algorithms demand computationally intensive data processing, which may involve data migration or the deployment of data analytics functions close to the data pool, such as within the same data center. In these instances, resource considerations such as network bandwidth for data migration or computational power for local data processing become paramount.

- *Data Representation:* Lastly, the need for data representation emerges prominently within the hierarchical NS management layers. As per the 3GPP guidelines, the NSSMF, NSMF, and CSMF must harmonize their operations [90]. Hence, it is rational to assume such interplay could be facilitated through mechanisms such as data abstraction, where NSSMF distills an abstracted knowledge of its systems, which is then presented to the NSMF. For example, NSSMFs could process the historical delay observed in their systems and share delay predictions with the NSMF. This abstraction enables the NSMF to craft an expanded end-to-end representation of the network slice, encompassing vital parameters such as end-to-end delays and total resource cost. The data representation should serve the optimization objectives of MO functions, enabling them to autonomously fine-tune the design and operation of network slices, as will be discussed in more detail in the next section.

  Key Insights: Limited research exists on the consideration of data representation within the NS management layers. Data representation could be facilitated through data abstraction mechanisms. For instance, data abstraction can be implemented using various techniques like feature selection or statistical summarization. NSSMFs could employ ML models to extract key performance patterns that capture essential service characteristics. These compressed representations enable efficient communication between management layers while preserving critical information for decision-making. Additionally, semantic abstraction using ontologies can help standardize how different domains represent their data, ensuring consistent interpretation across layers. The abstraction level could be dynamically adjusted based on the current network state and service requirements, e.g., during normal operation, coarse-grained metrics may suffice, while detailed data may be needed during service degradation or resource conflicts. To retrieve such detailed data when necessary, the framework can temporarily lower the abstraction level and query raw performance metrics directly from monitoring agents or telemetry systems. This can be achieved through on-demand data collection policies, fine-grained monitoring probes, or dynamic instrumentation techniques that selectively activate deeper metrics collection based on contextual triggers.

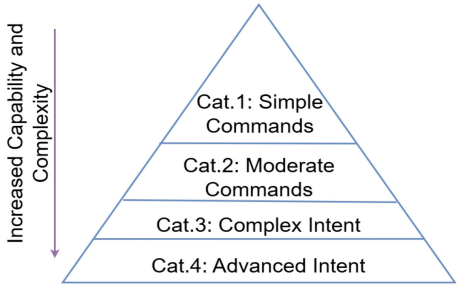


Fig. 3. Intent categories.

V. Future IoT Systems: Evolution of Services and Open Ecosystems

This section provides a forward-looking perspective on IoT systems, discussing the evolution of intent and NS terminologies that jointly enable future IoT services. Moreover, the section examines the IoT ecosystem.

A. The Role of Intent and Network Slicing

Within both the literature and industry realms, the precise definition of "intent" remains a subject of ongoing deliberation. A common source of confusion is the interchangeable use of "intent" and "network slice" in various contexts [38], [46]. As I-IBN systems become more prevalent, establishing clear distinctions between these terms becomes increasingly critical.

To address this need, we introduce the intent categories presented in Fig. 3, which serves as a framework illustrating the evolution of intent and its relation to NS. The first category encompasses simple command intents, exemplified by requests such as initiating a file transfer between two endpoints. Practical implementations of these intents are evident in smart chatbots and personal assistants like Amazon Alexa [91]. In this context, users, including smart home residents, can control home appliances through straightforward expressions such as "Alexa, turn on the T.V." Simple command intents require translating those intents to control commands that can be enabled by the underlying system.

Moving to the second category, we encounter moderate command intents that demand more intricate alterations to the system configuration. An instance could be a request for a secure connection between two endpoints [2]. In domains like eHealth, a corresponding intent might specify restricted access to patient records for doctors and nurses exclusively [92].

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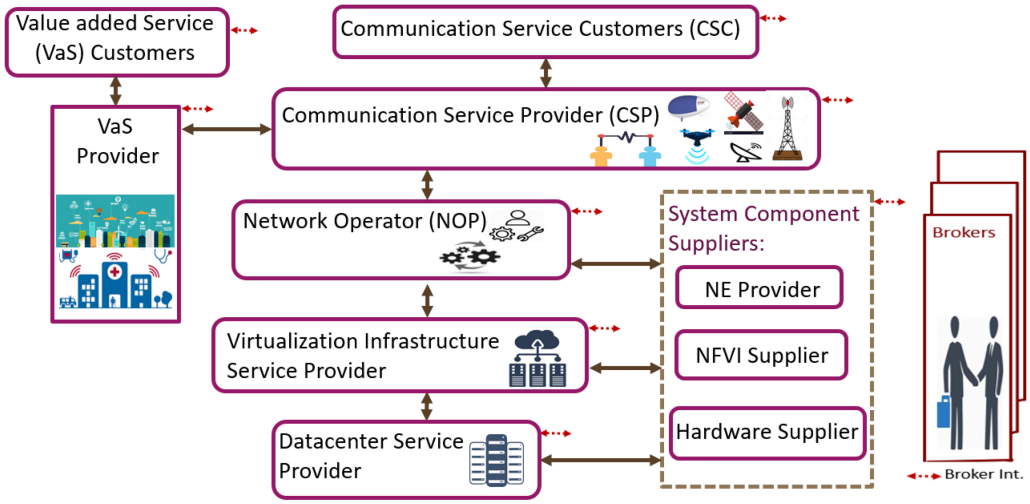


Fig. 4. The IoT ecosystems.

Fulfilling these intents mandates the enforcement of security measures, such as access control policies within the network domains specified by the intents.

The third category pertains to complex intents that necessitate substantial modifications to the system configurations, potentially involving the instantiation of a network slice. As an illustration, consider an intent within an eHealth system, such as "Establish a smart virtual assistant for the analysis of real-time patients' data." The heightened attention on chatbots and smart personal assistants in the eHealth domain underscores the significance of such endeavors [93], [94]. Realizing these advanced intents requires integrated capabilities, including medical sensors, data analytics, and access to servers hosting medical records. To fulfill the intended outcome of complex intents, I-IBN systems must instantiate a service, with the service instance traversing all I-IBN components.

Category four pertains to highly sophisticated intents that may necessitate tangible modifications to the foundational network infrastructure. To illustrate, consider a scenario where a homeowner aspires to implement intelligent surveillance in their backyard. However, the requisite surveillance cameras are not currently installed. In response, the I-IBN system must autonomously strategize and initiate a request for the installation of intelligent cameras in the designated backyard area. This process might involve communication and planning with multiple business entities. Furthermore, the I-IBNS system undertakes the task of establishing seamless communication channels between the recently integrated physical network elements–such as the smart cameras–and the pre-existing infrastructure, exemplified by the smart home IoT system. This intricate process showcases the I-IBNS system's capacity to dynamically orchestrate physical alterations while ensuring cohesive integration within the broader network framework.

Key Insights: Intent represents a general form of service request that can span any of the aforementioned categories with a broad spectrum of autonomous capabilities and complexities. A network slice request is a specific means to fulfill an intent. Importantly, intents falling into the second or third categories could potentially be translated into a network slice request or an NSaaS request. Moreover, network slice requests are typically formulated by Information Technology (IT) professionals, whereas intents may be expressed by both technical and non-technical professionals.

It is worth noting that most existing research focuses on simple command intents, primarily, enabling automation of basic technical tasks such as the work in [39], and moderate command intents enabled by NS, e.g., [49]. Nevertheless, the presented intent categories provide a forward-looking framework for the evolution of IoT services from simple commands to advanced intents.

B. IoT Ecosystem

Understanding the IoT ecosystem of the future is essential for addressing the challenges of implementing IBN in next-generation IoT systems. We present a reference ecosystem model that builds on virtualization business concepts and aligns with 3GPP's vision for future IoT systems. The reference ecosystem is depicted in Fig. 4, which shows the entities involved and the arrows denote potential interfaces for business relationships. The subsequent section will expound upon these entities and interfaces.

Value Added Service Provider (VaSp): VaSps, constituting application providers, encompass entities such as Over-the-Top (OTT) application providers and vertical industries, for instance, healthcare sectors providing innovative services like eHealth services to Value-added Service (VaS) customers.

Communication Service Provider (CSP): CSPs assume the role of offering communication services, such as WiFi and 5G coverage, to end-users, namely Communication Service Customers (CSC), or VaSp.

Network Operator (NOP): NOPs embody entities possessing expertise in the management, operation, maintenance, and planning of communication networks. Within NOPs, a noteworthy subclass is virtual network operators (VNOs), which share expertise in network management with NOPs but do not own physical networks. VNOs can be engaged by different entities to operate their networks.

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Virtual Infrastructure Service Provider (vInfP): vInfPs are instrumental in deploying and managing physical resources in the form of data center networks. They provide logical interfaces to customers, facilitating virtual connectivity of network resources. Examples include cloud providers and 5G system owners offering NSaaS models like RAN Slice as a Service (RANSaaS) or end-to-end 5G NSaaS.

Data Center Service Provider: This entity specializes in providing data center services, encompassing the deployment, management, and migration of data centers.

System Component Suppliers (SysComp): SysComp denotes manufacturing industries producing system building blocks, including network equipment (NE) providers offering Network Function (NF) implementations, and hardware suppliers like commodity server providers.

Brokers: Brokers represent entities that negotiate with multiple system owners who manage and deploy orchestration components such as NSMF, NSSMF, or NFVO to integrate customized solutions and subsequently offer them in the market. These interactions typically involve negotiating service requirements and coordinating the provisioning of customized solutions. However, in future I-IBNS systems with advanced intent capabilities, the role of brokers may evolve into autonomous negotiation and deployment agents powered by LLMs. In such scenarios, broker systems or LLM-based brokers will be capable of independently negotiating business agreements and translating them into actionable service deployment requests [95]. These broker LLMs will directly interface with orchestration systems, initiating slice instantiation and resource allocation processes without human intervention, thus enabling fully automated end-to-end service fulfillment across multi-domain IoT environments.

Fundamentally, the IoT ecosystem presents an array of implementation possibilities, drawing upon diverse technologies and business entities to effectively address the requirements of IoT services. The openness of this ecosystem is notably facilitated by NS technology, enabling the virtualization of entire network segments and thereby unlocking a spectrum of business models, exemplified by the Business to Business to Consumer (B2B2C) model [25]. This model, in turn, includes scenarios where VaSps can directly solicit NSaaS from CSPs to establish direct connections with their VaS customers. Moreover, brokers can offer customized technological solutions, resulting in wider implementation options with the potential to satisfy diverse IoT service requirements.

This departure from the traditional model denotes a paradigm shift wherein services are no longer exclusively under the control of CSPs, thus dismantling market monopolies. Various business entities gain prominence and wield influence within this dynamic landscape. In the illustrated scenario, VaS customers have the option to subscribe directly to VaSps without necessitating the intermediary involvement of CSPs. This decentralization of service provisioning introduces a more competitive and diversified market structure.

It is noteworthy that, within this conceptualization, organizational roles exhibit a fluidity, with entities concurrently undertaking multiple roles. For example, a Mobile Network Operator (MNO) can seamlessly function both as an NOP and a CSP, showcasing the adaptability and versatility inherent in this evolving IoT ecosystem.

Key Insights: Despite the advantages of the open IoT ecosystem, it introduces several challenges to the deployment of I-IBNS systems. Privacy preservation becomes critical as sensitive data could flow across multiple business entities. Also, MO coordination becomes complex when multiple service providers need to collaborate. This is particularly evident in scenarios where end-to-end service delivery requires orchestrating resources across different cloud providers, edge networks, and access networks, each potentially managed by different organizations.

The automation of network operations across this diverse ecosystem presents its own set of challenges. When different providers use varying automation frameworks and interfaces, ensuring seamless integration becomes difficult. For instance, automating the deployment of an end-to-end service might require coordinating across multiple providers' automation systems, each with its own intent translation mechanisms and policy enforcement frameworks. This heterogeneity requires clear coordination and communication between the systems.

Finally, business agreements and legislation pose challenges as service providers must navigate varying regulatory requirements across different regions, e.g., or countries, while establishing clear SLA and defining liability boundaries in cases of service disruption or security breaches.

VI. Open Issues and Research Directions

This section discusses several open research issues and future directions related to I-IBNS systems:

- *The Role of LLM in Future I-IBNS Systems:* Within I-IBNS systems, the evolution of LLMs can be critical in enabling network automation, network abstractions, and enhancing user experience. LLMs are increasingly utilized in intent translation systems, where users can articulate intents using natural language by communicating with intelligent chatbots enabled by LLMs. LLMs then convert these intents into various outputs such as NSDs [50], [96], network services (NS) [97], system configurations [98], [99], execution plans, or policies [100], [101]. The translation capability of an LLM depends not only on its proficiency in interpreting the user's unstructured natural language but also on its ability to understand system data, configurations, and policies. An emerging research avenue is leveraging LLMs for intent resolution. For example, the work in [102] introduces "Maestro," a system designed to enable conflict resolution and stakeholder collaboration in shared intent-based 6G networks. Maestro achieves this by abstracting diverse network infrastructures into declarative intents spanning business, service, and network planes. Another study [103] explores the comprehensive application of LLMs throughout the intent lifecycle, encompassing intent decomposition, translation, negotiation, activation, and assurance. This approach demonstrates the potential of natural language-driven configurations and management of network services.

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- *Defining IBN components within the NS Management Layers:* While numerous studies have investigated I-IBNS across various domains like RAN and core networks, there has been a notable absence of focus on NS management functions. For instance, the CSMF, NSMF, and NSSMF play pivotal roles in resource management within large-scale IoT systems. Yet, existing research has not explored how these functions can integrate IBN methodologies. There is a need for research that focuses on addressing the automation of tasks associated with each NS management function and exploring how these functions can effectively communicate with each other. This integration could potentially facilitate the realization of fully automated networks across all management levels.
- *Importance of Change Management in Network Automation:* As network automation becomes increasingly integrated into modern MO frameworks, a stressing need arises to understand the effect of changes in the network. This aspect is important to ensure that modifications made within a specific network domain do not have adverse effects on services in other domains. Additionally, there is a critical question of how configuration changes applied to one service can be executed without compromising the QoS of concurrently instantiated services. While the realm of IBN has initiated efforts to address the intent resolution component, there persists an imperative to adopt a holistic perspective. The introduction of a change management layer becomes pivotal in mitigating these challenges. This layer is designed to assess the impact of a configuration change within the network, continually analyze network data to discern interrelated systems with non-clear connections, and evaluate the effects across different domains. In deploying a new intent, the change management layer supports the intent resolution components by providing comprehensive data on the impact of the MO configurations.
- *Investigating the Security Implications of I-IBNS:* The security considerations within I-IBNS frameworks are of paramount significance, given the intricate nature of IoT systems and the integration of network automation solutions. The impact of such automation on the overall security of the system remains unclear. For instance, adversaries with an understanding of the employed automation mechanism could influence the system to take specific actions. Such scenarios should be carefully considered. Furthermore, despite existing research on the security of MANO NFV orchestrators, the security aspects of the broader I-IBNS framework have yet to be comprehensively addressed [68]. An effective security framework for I-IBNS should inherently incorporate well-defined security policies. These policies should be integral to the overarching framework, encompassing communication protocols among various MO entities.
- *Enabling Intent-based System-to-System Communications in Multi-Administrator Domains* An interesting research avenue is to extend the concept of IBN to system-to-system communications where systems could be governed by distinct business entities. This application can facilitate the realization of an advanced intent, which is the intent category four, wherein business-to-business negotiations and contracts can transpire without direct engagement from the I-IBNS users. A business intent module within the I-IBNS system holds the capability to engage in negotiations with other business intent modules independently of direct involvement from the system owner. System owners maintain business intents such as monetary cost limits, accordingly, I-IBNS systems foster seamless business communications enabling advanced intents.

VII. Conclusion

This paper surveys I-IBNS systems, instrumental facilitators for next-generation IoT systems that integrate NS and IBN technologies. I-IBNS systems rely on the foundational frameworks of data management and resource management. The former ensures the secure and operational incorporation of big data generated by complex underlying IoT systems into the MO entities, thereby establishing an intelligent MO framework. The latter allows for efficient MO of intent-oriented network slices. The paper systematically explores both areas, delving into associated challenges and presenting avenues for further research. Moreover, the paper presents a forward looking perspective on the role of intent and NS and the IoT ecosystem.

References

[1] A. Clemm, L. Ciavaglia, L. Z. Granville, and J. Tantsura, "Intent-based networking—Concepts and definitions," IETF, RFC 9315, Oct. 2022.
[2] A. Leivadeas and M. Falkner, "A survey on intent-based networking," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 1, pp. 625–655, 1st Quart., 2023.
[3] A. Leivadeas and M. Falkner, "Autonomous network assurance in intent based networking: Vision and challenges," in *Proc. 32nd Int. Conf. Comput. Commun. Netw. (ICCCN)*, 2023, pp. 1–10.
[4] K. Abbas, T. A. Khan, M. Afaq, and W.-C. Song, "Network slice lifecycle management for 5G mobile networks: An intent-based networking approach," *IEEE Access*, vol. 9, pp. 80128–80146, 2021.
[5] M. Xie, P. H. Gomes, J. Niemöller, and J. P. Waldemar, "Intent-driven management for multi-vertical end-to-end network slicing services," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, 2022, pp. 1285–1291.
[6] S. Wijethilaka and M. Liyanage, "Survey on network slicing for Internet of Things realization in 5G networks," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 2, pp. 957–994, 2nd Quart., 2021.
[7] C. Yang, X. Mi, Y. Ouyang, R. Dong, J. Guo, and M. Guizani, "Smart intent-driven network management," *IEEE Commun. Mag.*, vol. 61, no. 1, pp. 106–112, Jan. 2023.
[8] J. Zhang, C. Yang, A. Anpalagan, R. Dong, and L. Zhang, "IDSM: Intent-driven slice management and maintenance for 6G RANs," in *Proc. IEEE 22nd Int. Conf. Commun. Technol. (ICCT)*, 2022, pp. 1244–1249.
[9] N. F. Cheng, T. Pamuklu, and M. Erol-Kantarci, "Reinforcement learning based resource allocation for network slices in O-RAN midhaul," in *Proc. IEEE 20th Consum. Commun. Netw. Conf. (CCNC)*, 2023, pp. 140–145.
[10] J. Zhang et al., "Intent-driven RAN slice orchestration: A multi-agent deep reinforcement learning based approach," in *Proc. IEEE/CIC Int. Conf. Commun. China (ICCC Workshops)*, 2023, pp. 1–6.
[11] L. Pang, C. Yang, D. Chen, Y. Song, and M. Guizani, "A survey on intent-driven networks," *IEEE Access*, vol. 8, pp. 22862–22873, 2020.
[12] E. Zeydan and Y. Turk, "Recent advances in intent-based networking: A survey," in *Proc. IEEE 91st Veh. Technol. Conf. (VTC-Spring)*, 2020, pp. 1–5.

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[13] Y. Ouyang, C. Yang, Y. Song, X. Mi, and M. Guizani, "A brief survey and implementation on refinement for intent-driven networking," *IEEE Netw.*, vol. 35, no. 6, pp. 75–83, Nov./Dec. 2021.

[14] M. Chahbar, G. Diaz, A. Dandoush, C. Cérin, and K. Ghoumid, "A comprehensive survey on the E2E 5G network slicing model," *IEEE Trans. Netw. Service Manag.*, vol. 18, no. 1, pp. 49–62, Mar. 2021.

[15] S. D. A. Shah, M. A. Gregory, and S. Li, "Cloud-native network slicing using software defined networking based multi-access edge computing: A survey," *IEEE Access*, vol. 9, pp. 10903–10924, 2021.

[16] H. H. Esmat, B. Lorenzo, and W. Shi, "Toward resilient network slicing for satellite–terrestrial edge computing IoT," *IEEE Internet Things J.*, vol. 10, no. 16, pp. 14621–14645, Aug. 2023.

[17] Y. Wu, H.-N. Dai, H. Wang, Z. Xiong, and S. Guo, "A survey of intelligent network slicing management for Industrial IoT: Integrated approaches for smart transportation, smart energy, and smart factory," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 2, pp. 1175–1211, 2nd Quart., 2022.

[18] H. P. Phyu, D. Naboulsi, and R. Stanica, "Machine learning in network slicing—A survey," *IEEE Access*, vol. 11, pp. 39123–39153, 2023.

[19] N. Saha, M. Zangooei, M. Golkarifard, and R. Boutaba, "Deep reinforcement learning approaches to network slice scaling and placement: A survey," *IEEE Commun. Mag.*, vol. 61, no. 2, pp. 82–87, Feb. 2023.

[20] A. Donatti et al., "Survey on machine learning-enabled network slicing: Covering the entire life cycle," *IEEE Trans. Netw. Service Manag.*, vol. 21, no. 1, pp. 994–1011, Feb. 2024.

[21] M. A. Habibi et al., "Toward an open, intelligent, and end-to-end architectural framework for network slicing in 6g communication systems," *IEEE Open J. Commun. Soc.*, vol. 4, pp. 1615–1658, 2023.

[22] P. J. Horn. "Autonomic computing: IBM's perspective on the state of information technology." 2001. [Online]. Available: https://www.scirp.org/reference/referencespapers?referenceid=1675401

[23] W. S. Liu et al., "Policy-based management framework for the simplified use of policy abstractions (SUPA)," IETF, RFC 8328, Mar. 2018.

[24] D. E. Sarmiento, A. Lebre, L. Nussbaum, and A. Chari, "Decentralized SDN control plane for a distributed cloud-edge infrastructure: A survey," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 1, pp. 256–281, 1st Quart., 2021.

[25] "Telecommunication management; study on management and orchestration of network slicing for next generation network," 3GPP, Sophia Antipolis, France, Rep. 28.801, 2018.

[26] T. Zhang, H. Qiu, L. Linguaglossa, W. Cerroni, and P. Giaccone, "NFV platforms: Taxonomy, design choices and future challenges," *IEEE Trans. Netw. Service Manag.*, vol. 18, no. 1, pp. 30–48, Mar. 2021.

[27] *The European Telecommunications Standards Institute Industry Specification Group on Network Function Virtualization (ETSI ISG NFV)*, ETSI, Sophia Antipolis, France, 2019.

[28] S. Duan et al., "Distributed artificial intelligence empowered by end-edge-cloud computing: A survey," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 1, pp. 591–624, 1st Quart., 2023.

[29] "Multi-access edge computing (MEC), MEC management, part 2: Application lifecycle, rules and requirements management," ETSI, Sophia Antipolis, France, Rep. ETSI GS MEC 010-2, 2022.

[30] M. Mukherjee, L. Shu, and D. Wang, "Survey of fog computing: Fundamental, network applications, and research challenges," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 3, pp. 1826–1857, 3rd Quart., 2018.

[31] M. Vaezi et al., "Cellular, wide-area, and non-terrestrial IoT: A survey on 5G advances and the road toward 6G," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 2, pp. 1117–1174, 2nd Quart., 2022.

[32] M. A. Hossain, A. R. Hossain, W. Liu, N. Ansari, A. Kiani, and T. Saboorian, "A distributed collaborative learning approach in 5G+ core networks," *IEEE Netw.*, vol. 38, no. 1, pp. 288–295, Jan. 2024.

[33] S. Sevgican, M. Turan, K. Gökarslan, H. B. Yilmaz, and T. Tugcu, "Intelligent network data analytics function in 5G cellular networks using machine learning," *J. Commun. Netw.*, vol. 22, no. 3, pp. 269–280, 2020.

[34] M. Polese, L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "Understanding O-RAN: Architecture, interfaces, algorithms, security, and research challenges," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 2, pp. 1376–1411, 2nd Quart., 2023.

[35] T. A. Khan, A. Mehmood, J. J. D. Ravera, A. Muhammad, K. Abbas, and W.-C. Song, "Intent-based orchestration of network slices and resource assurance using machine learning," in *Proc. IEEE/IFIP Netw. Oper. Manag. Symp. (NOMS)*, 2020, pp. 1–2.

[36] T. A. Khan, K. Abbass, A. Rafique, A. Muhammad, and W.-C. Song, "Generic intent-based networking platform for E2E network slice orchestration & lifecycle management," in *Proc. 21st Asia–Pac. Netw. Oper. Manag. Symp. (APNOMS)*, 2020, pp. 49–54.

[37] K. Abbas, T. A. Khan, M. Afaq, J. J. D. Rivera, and W.-C. Song, "Network data analytics function for IBN-based network slice lifecycle management," in *Proc. 22nd Asia–Pac. Netw. Oper. Manag. Symp. (APNOMS)*, 2021, pp. 148–153.

[38] M. Gharbaoui, B. Martini, and P. Castoldi, "Implementation of an intent layer for SDN-enabled and QoS-aware network slicing," in *Proc. IEEE 7th Int. Conf. Netw. Softw. (NetSoft)*, 2021, pp. 17–23.

[39] K. Mehmood, H. V. K. Mendis, K. Kralevska, and P. E. Heegaard, "Intent-based network management and orchestration for smart distribution grids," in *Proc. 28th Int. Conf. Telecommun. (ICT)*, 2021, pp. 1–6.

[40] R. Perez, A. Zabala, and A. Banchs, "ALVIU: An intent-based SD-WAN orchestrator of network slices for enterprise networks," in *Proc. IEEE 7th Int. Conf. Netw. Softw. (NetSoft)*, 2021, pp. 211–215.

[41] D. Wang, R. Su, and S. Zhang, "An intent-based smart slicing framework for vertical industry in B5G networks," in *Proc. IEEE/CIC Int. Conf. Commun. China (ICCC Workshops)*, 2021, pp. 389–394.

[42] N. Gritli, F. Khendek, and M. Toeroe, "Network slice provisioning taking into account tenant intents and operator policies," in *Proc. IEEE Conf. Netw. Function Virtual. Softw. Defined Netw. (NFV-SDN)*, 2021, pp. 167–173.

[43] N. F. S. D. Sousa and C. E. Rothenberg, "CLARA: Closed loop-based zero-touch network management framework," in *Proc. IEEE Conf. Netw. Function Virtualization Softw. Defined Netw. (NFV-SDN)*, 2021, pp. 110–115.

[44] E. Chirivella-Perez, P. Salva-Garcia, R. Ricart-Sanchez, J. A. Calero, and Q. Wang, "Intent-based E2E network slice management for industry 4.0," in *Proc. Joint Eur. Conf. Netw. Commun. 6G Summit (EuCNC/6G Summit)*, 2021, pp. 353–358.

[45] C. Fernández et al., "Application of multi-pronged monitoring and intent-based networking to verticals in self-organizing networks," in *Proc. 5th Int. Conf. Adv. Commun. Technol. Netw. (CommNet)*, 2022, pp. 1–10.

[46] K. Abbas, T. A. Khan, M. Afaq, and W.-C. Song, "Ensemble learning-based network data analytics for network slice orchestration and management: An intent-based networking mechanism," in *Proc. IEEE/IFIP Netw. Oper. Manag. Symp. (NOMS)*, 2022, pp. 1–5.

[47] K. Abbas et al., "AI-driven data analytics and intent-based networking for orchestration and control of B5G consumer electronics services," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 2155–2169, Feb. 2024.

[48] J. Mcnamara et al., "NLP powered intent based network management for private 5G networks," *IEEE Access*, vol. 11, pp. 36642–36657, 2023.

[49] J. Zhang et al., "Intent-driven closed-loop control and management framework for 6G open RAN," *IEEE Internet Things J.*, vol. 11, no. 4, pp. 6314–6327, Feb. 2024.

[50] A. Mekrache and A. Ksentini, "LLM-enabled intent-driven service configuration for next generation networks," in *Proc. IEEE 10th Int. Conf. Netw. Softw. (NetSoft)*, 2024, pp. 253–257.

[51] D. H. Hussein and M. Ibnkahla, "Privacy-preserving intelligent intent-based network slicing for IoT systems," *IEEE Internet Things J.*, early access, Apr. 24, 2025, doi: 10.1109/JIOT.2025.3564138.

[52] D. H. Hussein and M. Ibnkahla, "Intelligent and autonomous edge slicing for IoT systems," *IEEE Internet Things J.*, early access, Feb. 26, 2025, doi: 10.1109/JIOT.2025.3546124.

[53] J. Bin, Z. Bahrami, C. A. Rahman, S. Du, S. Rogers, and Z. Liu, "Foreground fusion-based liquefied natural gas leak detection framework from surveillance thermal imaging," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 7, no. 4, pp. 1151–1162, Aug. 2023.

[54] I. AlShiab, A. Leivadeas, and M. Ibnkahla, "Virtual sensing networks and dynamic rpl-based routing for IoT sensing services," in *Proc. IEEE Int. Conf. Commun. (ICC)*, 2021, pp. 1–6.

[55] A. Gharib and M. Ibnkahla, "User security oriented information-centric IoT nodes clustering with graph convolution networks," *IEEE Internet Things J.*, vol. 11, no. 5, pp. 8311–8326, Mar. 2024.

[56] A. Alwarafy, K. A. Al-Thelaya, M. Abdallah, J. Schneider, and M. Hamdi, "A survey on security and privacy issues in edge-computing-assisted Internet of Things," *IEEE Internet Things J.*, vol. 8, no. 6, pp. 4004–4022, Mar. 2021.

[57] A. Jarwan, A. Sabbah, and M. Ibnkahla, "Information-oriented traffic management for energy-efficient and loss-resilient IoT systems," *IEEE Internet Things J.*, vol. 9, no. 10, pp. 7388–7403, May 2022.

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[58] A. Gharib and M. Ibnkahla, "Heterogeneous cluster-based information-centric sensor networks with user security satisfaction," *IEEE Internet Things J.*, vol. 10, no. 9, pp. 8123–8139, May 2023.

[59] A. Jarwan and M. Ibnkahla, "Edge-based federated deep reinforcement learning for IoT traffic management," *IEEE Internet Things J.*, vol. 10, no. 5, pp. 3799–3813, Mar. 2023.

[60] W. Yao, H. Zhao, and H. Shi, "Privacy-preserving collaborative intrusion detection in edge of Internet of Things: A robust and efficient deep generative learning approach," *IEEE Internet Things J.*, vol. 11, no. 9, pp. 15704–15722, May 2024.

[61] M. A. Issa, M. Ibnkahla, A. Matrawy, and A. Eldosouky, "Temporal partitioned federated learning for IoT intrusion detection systems," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, 2024, pp. 1–6.

[62] A. Farhat, A. Eldosouky, M. Ibnkahla, and A. Matrawy, "Interaction-aware trust management scheme for iot systems with machine learning-based attack detection," *IEEE Internet Things J.*, early access, Feb. 6, 2025, doi: 10.1109/JIOT.2025.3539646.

[63] Z. Liao, X. Pang, J. Zhang, B. Xiong, and J. Wang, "Blockchain on security and forensics management in edge computing for IoT: A comprehensive survey," *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 2, pp. 1159–1175, Jun. 2022.

[64] D. H. Hussein, E. Houlahan, A. Janelle-Goode, T. Lumsden, and M. Ibnkahla, "A blockchain-based dual identity management and authentication framework for IoT networks," in *Proc. IEEE 10th World Forum Internet Things (WF-IoT)*, 2024, pp. 544–549.

[65] D. H. Hussein, R. Anbarasu, A. Matrawy, and M. Ibnkahla, "Towards a decentralized access control system for IoT platforms based on blockchain technology," in *Proc. Int. Symp. Netw. Comput. Commun. (ISNCC)*, 2020, pp. 1–6.

[66] M. Devi and A. Majumder, "Side-channel attack in Internet of Things: A survey," in *Applications of Internet of Things*. Singapore: Springer, 2021, pp. 213–222.

[67] K. Cao et al., "Enhancing physical-layer security for IoT with nonorthogonal multiple access assisted semi-grant-free transmission," *IEEE Internet Things J.*, vol. 9, no. 24, pp. 24669–24681, Dec. 2022.

[68] C. D. Alwis, P. Porambage, K. Dev, T. R. Gadekallu, and M. Liyanage, "A survey on network slicing security: Attacks, challenges, solutions and research directions," *IEEE Commun. Surveys Tuts.*, vol. 26, no. 1, pp. 534–570, 1st Quart., 2024.

[69] D. Sattar and A. Matrawy, "Proactive and dynamic slice allocation in sliced 5G core networks," in *Proc. Int. Symp. Netw. Comput. Commun. (ISNCC)*, 2020, pp. 1–8.

[70] R. Song, B. Xiao, Y. Song, S. Guo, and Y. Yang, "A survey of blockchain-based schemes for data sharing and exchange," *IEEE Trans. Big Data*, vol. 9, no. 6, pp. 1477–1495, Dec. 2023.

[71] E. Zeydan, J. Baranda, J. Mangues-Bafalluy, and Y. Turk, "Blockchain for network service orchestration: Trust and adoption in multi-domain environments," *IEEE Commun. Stand. Mag.*, vol. 7, no. 2, pp. 16–22, Jun. 2023.

[72] E. Zeydan, J. Baranda, J. Mangues-Bafalluy, Y. Turk, and S. B. Ozturk, "Blockchain-based service orchestration for 5G vertical industries in multicloud environment," *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 4, pp. 4888–4904, Dec. 2022.

[73] D. H. Hussein and M. Ibnkahla, "An IoT traffic modeling framework and its application to autonomous edge scaling," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, 2022, pp. 5656–5661.

[74] D. H. Hussein and M. Ibnkahla, "A novel mathematical framework for modeling application-specific IoT traffic," *IEEE Internet Things J.*, vol. 11, no. 2, pp. 2364–2381, Jan. 2024.

[75] M. I. S. Oliveira and B. F. Lóscio, "What is a data ecosystem?" in *Proc. 19th Annu. Int. Conf. Digit. Govt. Res. Data Age*, 2018, p. 18.

[76] Z.-Y. Shae and J. J. Tsai, "On the design of medical data ecosystem for improving healthcare research and commercial incentive," in *Proc. IEEE 3rd Int. Conf. Cogn. Mach. Intell. (CogMI)*, 2021, pp. 124–131.

[77] M. Franklin, A. Halevy, and D. Maier, "From databases to dataspaces: A new abstraction for information management," *SIGMOD Rec.*, vol. 34, pp. 27–33, Dec. 2005.

[78] J. Möller, D. Jankowski, A. Lamm, and A. Hahn, "Data management architecture for service-oriented maritime testbeds," *IEEE Open J. Intell. Transp. Syst.*, vol. 3, pp. 631–649, 2022.

[79] H. Zhang, H. Zhou, and M. Erol-Kantarci, "Federated deep reinforcement learning for resource allocation in O-RAN slicing," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, 2022, pp. 958–963.

[80] H. Zhang et al., "On-device intelligence for 5G RAN: Knowledge transfer and federated learning enabled UE-centric traffic steering," *IEEE Trans. Cogn. Commun. Netw.*, vol. 10, no. 2, pp. 689–705, Apr. 2024.

[81] H. Zhou, M. Erol-Kantarci, and H. V. Poor, "Knowledge transfer and reuse: A case study of AI-enabled resource management in RAN slicing," *IEEE Wireless Commun.*, vol. 30, no. 5, pp. 160–169, Oct. 2023.

[82] F. Z. Yousaf, V. Sciancalepore, M. Liebsch, and X. Costa-Perez, "MANOaaS: A multi-tenant NFV MANO for 5G network slices," *IEEE Commun. Mag.*, vol. 57, no. 5, pp. 103–109, May 2019.

[83] N. Slamnik-Kriještorac, G. M. Yilma, M. Liebsch, F. Z. Yousaf, and J. M. Marquez-Barja, "Collaborative orchestration of multi-domain edges from a connected, cooperative and automated mobility (CCAM) perspective," *IEEE Trans. Mobile Comput.*, vol. 22, no. 4, pp. 2001–2020, Apr. 2023.

[84] J. S. Choi, S. J. Chun, and S. Lee, "Hierarchical distributed overarching architecture of decoupled federation and orchestration frameworks for multidomain NFV MANOS," *IEEE Commun. Mag.*, vol. 60, no. 9, pp. 68–74, Sep. 2022.

[85] M. Abughazala and H. Muccini, "Modeling data analytics architecture for IoT applications using DAT," in *Proc. IEEE 20th Int. Conf. Softw. Architect. Companion (ICSA-C)*, 2023, pp. 284–291.

[86] A. Hutterer and B. Krumay. "Integrating heterogeneous data in dataspaces—A systematic mapping study." 2022. [Online]. Available: https://aisel.aisnet.org/pacis2022/222/

[87] A. Randles, D. O'Sullivan, J. Keeney, and L. Fallon, "Ontology driven closed control loop automation," in *Proc. IEEE 9th Int. Conf. Netw. Softw. (NetSoft)*, 2023, pp. 528–533.

[88] X. Chang et al., "KID: Knowledge graph-enabled intent-driven network with digital twin," in *Proc. 27th Asia–Pac. Conf. Commun. (APCC)*, 2022, pp. 272–277.

[89] V. Pupezescu, M.-C. Pupezescu, and L.-A. Perioară, "Optimizations of database management systems for real time IoT edge applications," in *Proc. 23rd Int. Carpathian Control Conf. (ICCC)*, 2022, pp. 171–176.

[90] "Study on RAN improvements for machine-type communications," 3GPP, Sophia Antipolis, France, Rep. 37.868, 2012.

[91] J. S. Edu, X. Ferrer-Aran, J. Such, and G. Suarez-Tangil, "SkillVet: Automated traceability analysis of Amazon Alexa skills," *IEEE Trans. Depend. Secure Comput.*, vol. 20, no. 1, pp. 161–175, Feb. 2023.

[92] Y. Njah, A. Leivadeas, J. Violos, and M. Falkner, "Towards intent-based network automation for smart environments: A healthcare 4.0 use case," *IEEE Access*, vol. 11, pp. 136565–136576, 2023.

[93] J. Qiu et al., "Large AI models in health informatics: Applications, challenges, and the future," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 12, pp. 6074–6087, Dec. 2023.

[94] D. Sharma, S. Kaushal, H. Kumar, and S. Gainder, "ChatBots in healthcare: Challenges, technologies and applications," in *Proc. 4th Int. Conf. Artif. Intell. Speech Technol. (AIST)*, 2022, pp. 1–6.

[95] D. Krishnamurthy and V. Neelanath, "Establishing a robust LLMOPs framework for intelligent automation: Strategies and best practices," in *Proc. Emerg. Technol. Intell. Syst. (ETIS)*, 2025, pp. 1–5.

[96] D. M. Manias, A. Chouman, and A. Shami, "Towards intent-based network management: Large language models for intent extraction in 5G core networks," in *Proc. 20th Int. Conf. Design Rel. Commun. Netw. (DRCN)*, 2024, pp. 1–6.

[97] D. Adanza et al., "IntentLLM: An Ai ChatBot to create, find, and explain slice intents in TeraflowSDN," in *Proc. IEEE 10th Int. Conf. Netw. Softw. (NetSoft)*, 2024, pp. 307–309.

[98] S. D'urso, M. Fontana, B. Martini, and F. Sciarrone, "Enhancing intent acquisition and translation with large language models and intelligent ChatBots: A DHCP use case," in *Proc. IEEE 10th Int. Conf. Netw. Softw. (NetSoft)*, 2024, pp. 19–24.

[99] O. G. Lira, O. M. Caicedo, and N. L. S. d. Fonseca, "Large language models for zero touch network configuration management," *IEEE Commun. Mag.*, vol. 6, no. 4, pp. 227–231, Dec. 2024.

[100] J. Lin, K. Dzeparoska, A. Tizghadam, and A. Leon-Garcia, "AppleSeed: Intent-based multi-domain infrastructure management via few-shot learning," in *Proc. IEEE 9th Int. Conf. Netw. Softw. (NetSoft)*, 2023, pp. 539–544.

[101] K. Dzeparoska, J. Lin, A. Tizghadam, and A. Leon-Garcia, "LLM-based policy generation for intent-based management of applications," in *Proc. 19th Int. Conf. Netw. Service Manag. (CNSM)*, 2023, pp. 1–7.

[102] I. Chatzistefanidis, A. Leone, and N. Nikaein, "Maestro: LLM-driven collaborative automation of intent-based 6G networks," *IEEE Netw. Lett.*, vol. 6, no. 4, pp. 227–231, Dec. 2024.

[103] A. Mekrache, A. Ksentini, and C. Verikoukis, "Intent-based management of next-generation networks: An LLM-centric approach," *IEEE Netw.*, vol. 38, no. 5, pp. 29–36, Sep. 2024.

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